Stat 134: Conceptual midterm review solutions

1 What distribution should I use?

Question 1

The number of sixes in 15 rolls of a fair six-sided die:

This is a Binomial$(15, 1/6)$ distribution because:

- We have a fixed number of trials (15 dice rolls)
- Each trial is independent
- The probability of success (rolling a six) is constant at $p = 1/6$
- The possible values are $\{0, 1, 2, \ldots, 15\}$, matching the binomial distribution's range

Question 2

The number of games of Roulette I must play until I win three times, if I only ever bet on black:

This is a Negative Binomial$(3, 18/38)$ distribution because:

- We count the number of trials until a fixed number of successes (3 wins)
- Each trial is independent
- The probability of success (winning by betting on black) is constant at $p = 18/38$
- The possible values are $\{3, 4, 5, \ldots\}$, matching the negative binomial's range

Question 3

The number of Aces in 5 cards drawn from a standard deck:

This is a Hypergeometric$(5, 52, 4)$ distribution because:

- We sample without replacement from a finite population (52 cards)
- The population has a known number of successes (4 Aces)

- We draw a fixed number of items (5 cards)
- The possible values are $\{0, 1, 2, 3, 4\}$, matching the hypergeometric distribution's range

Question 4

Balls are drawn with replacement from a box with 5 red balls and 45 non-red balls until a red ball is drawn. The number of balls drawn from this box:

This is a Geometric(5/50) distribution because:

- We count the number of trials until the first success
- Each trial is independent (sampling with replacement)
- The probability of success (drawing a red ball) is constant at $p = 5/50 = 1/10$
- The possible values are $\{1, 2, 3, \ldots\}$, matching the geometric distribution's range

Question 5

Balls are drawn with replacement from a box with 20 blue balls and 30 non-blue balls until a blue ball is drawn. The number of non-blue balls drawn from this box:

This is a Geometric(20/50) distribution on $\{0, 1, 2, \ldots\}$ because:

- We count the number of failures before the first success
- Each trial is independent (sampling with replacement)
- The probability of success (drawing a blue ball) is constant at $p = 20/50 = 2/5$
- The possible values are $\{0, 1, 2, \ldots\}$, matching the geometric distribution's range on failures before first success

Note: This differs from the previous problem because we're counting the number of non-blue balls (failures) rather than the total number of draws.

Question 6

Customers arrive at a checkout counter randomly, at a rate of 10 per hour. The number of customers that arrive at the checkout counter in 30 minutes:

This is a Poisson(5) distribution because:

- Events occur randomly and independently in a fixed interval (30 minutes)
- The rate of occurrence is constant (10 per hour = 5 per 30 minutes)
- The possible values are $\{0, 1, 2, \ldots\}$, matching the Poisson distribution's range

Solution 7

A fair 20-sided die is rolled 50 times. The face of the die shows a number greater than 15 in 30 of the rolls. The number of sixes in the 50 rolls:

This is a Binomial$(20, 1/20)$ distribution because:

- This is a conditional distribution since we are given that 30 of the rolls have a face greater than 15. Lets assume it is the last 30 rolls. This distribution takes values 0-20.
- The chance it is 2, P(2) is given by Baye's rule as P(there are 2 sixes in first 20)P(the last 30 rolls are greater than 15)/P(the last 30 rolls are greater than 15)=P(there are 2 sixes in first 20). This is $\binom{20}{2}(1/20)^2(19/20)^{18}$ which is the binomial formula for $n = 20, p = 1/20$. Hence we have $Bin(20, 1/20)$.
- Note that we only know that the last 30 rolls are greater than 15 but nothing about the first 20 rolls.

2 Do you have any recommended problems?

Question 1

Below, we denote B_i and R_i for $i = 1, 2$ as the event that the i-th marble is blue or red, respectively.

Question 1 - (i)

$$\begin{aligned}
&\mathrm{P}\Big(\text{the second marble} = \text{red}\Big) = \mathrm{P}(R_2) = \mathrm{P}(R_1, R_2) + \mathrm{P}(B_1, R_2) \\
=&\mathrm{P}\Big(\text{the first marble} = \text{red}, \text{the second marble} = \text{red}\Big) + \\
&\mathrm{P}\Big(\text{the first marble} = \text{blue}, \text{the second marble} = \text{red}\Big) \\
=&\frac{r}{r+b} \times \frac{r+m}{r+b+m} + \frac{b}{r+b} \times \frac{r}{r+b+m}.
\end{aligned}$$

Question 1 - (ii)

Using Bayes' theorem, we have

$$\begin{aligned}
\mathrm{P}(B_1|R_2) =& \frac{\mathrm{P}(R_2|B_1)\mathrm{P}(B_1)}{\mathrm{P}(R_2)} \\
=& \frac{\frac{r}{r+b+m} \times \frac{b}{r+b}}{\frac{r}{r+b} \times \frac{r+m}{r+b+m} + \frac{b}{r+b} \times \frac{r}{r+b+m}} \\
=& \frac{b}{r+m+b}.
\end{aligned}$$

Question 2

Question 2 - (i)

Using the complement principle, we have

$$\begin{aligned}\mathrm{P}\Big(\text{at least one six}\Big) =& 1-\mathrm{P}\Big(\text{no sixes}\Big)\\ =& 1-\Big(\frac{19}{20}\Big)^{10}.\end{aligned}$$

Question 2 - (ii)

Using the inclusion-exclusion principle, we have

$$\begin{aligned}\mathrm{P}\Big(\text{at least one five and one six}\Big) =& 1-\mathrm{P}\Big(\text{no fives}\Big)-\mathrm{P}\Big(\text{no sixes}\Big)+\mathrm{P}\Big(\text{no fives and no sixes}\Big)\\ =& 1-2\times\Big(\frac{19}{20}\Big)^{10}+\Big(\frac{18}{20}\Big)^{10}.\end{aligned}$$

Question 2 - (iii)

Since no number greater than 12 appears, we effectively roll a 12-sided die instead of a 20-sided die, where each face from 1 to 12 has an equal probability of $\frac{1}{12}$. We model this scenario as a binomial experiment, where a success is defined as rolling a six. The probability of rolling a six is $\frac{1}{12}$, and the number of trials is 10. Thus,

$$\mathrm{P}\Big(3 \text{ sixes given that no number larger than 12 appears}\Big) = \binom{10}{3}\Big(\frac{1}{12}\Big)^3\Big(\frac{11}{12}\Big)^7$$

Question 2 - (iv)

Using the multinomial formula, we have

$$\mathrm{P} = \frac{10!}{3!2!4!1!}\times\Big(\frac{1}{20}\Big)^{3+2+4+1} = \frac{10!}{3!2!4!1!}\times\Big(\frac{1}{20}\Big)^{10}$$

Question 2 - (v)

This is another multinomial probability:

- Choose 5 distinct numbers from 20: $\binom{20}{5}$.
- Assign counts to numbers: $\frac{10!}{3!2!2!2!1!}$.
- Probability of rolling any specific numbers: $\Big(\frac{1}{20}\Big)^{10}$.

Therefore, we have

$$\mathrm{P} = \binom{20}{5}\times\frac{10!}{3!2!2!2!1!}\times\Big(\frac{1}{20}\Big)^{10}$$

Question 3

Each die roll is independent, and the probability of rolling a six is $p = \frac{1}{6}$. Since we roll the die 1200 times, the number of sixes X follows a binomial distribution $X \sim \text{Bin}(n = 1200, p = \frac{1}{6})$. The mean and standard deviation of a binomial distribution are:

$$\mu = np = 1200 \times \frac{1}{6} = 200$$
$$\sigma = \sqrt{np(1-p)} = \sqrt{1200 \times \frac{1}{6} \times \frac{5}{6}} = \sqrt{166.67} \approx 12.91$$

Using the normal approximation, we approximate $X \approx N\left(200, 12.91^2\right)$.

Question 3 - (i)

Using the normal approximation, we apply the continuity correction:

$$\text{P}(X > 400) \approx \text{P}(Z > \frac{400.5 - 200}{12.91}) = \text{P}(Z > \frac{200.5}{12.91}) = \text{P}(Z > 15.53)$$

Question 3 - (ii)

Using the normal approximation with continuity correction:

$$\text{P}\Big(\frac{200 - m - 0.5 - 200}{12.91} \leq Z \leq \frac{200 + m + 0.5 - 200}{12.91}\Big) \approx 0.95$$
$$\implies \quad \text{P}\Big(\frac{-m - 0.5}{12.91} \leq Z \leq \frac{m + 0.5}{12.91}\Big) \approx 0.95$$

From standard normal tables, the z-score corresponding to a 95% probability range is $Z \approx \pm 1.96$. Thus,

$$\frac{m + 0.5}{12.91} = 1.96 \quad \implies \quad m \approx 24.8$$

Therefore, the approximate value of m is 25.

Question 4

Let T be the number of rolls until we get a one in a single experiment. Then T follows a geometric distribution with success probability $p = \frac{1}{n}$:

$$\text{P}(T = k) = (1-p)^{k-1}p \quad \text{and} \quad \text{P}(T > k) = (1-p)^k.$$

Define

- $X = \min\left(T_1, T_2, \cdots, T_{10}\right)$ (minimum number of rolls),
- $Y = \max\left(T_1, T_2, \cdots, T_{10}\right)$ (maximum number of rolls).

Question 4 - (i)

$$\begin{aligned}\mathrm{P}(X > k) =& \mathrm{P}\big(T_1 > k, T_2 > k, \ldots, T_{10} > k\big) \\ =& \big\{\mathrm{P}(T > k)\big\}^{10} && \text{(Independency)} \\ =& \big\{(1-p)^k\big\}^{10} = (1-p)^{10k}.\end{aligned}$$

Question 4 - (ii)

$$\begin{aligned}\mathrm{P}(Y > k) = 1 - \mathrm{P}(Y \le k) =& 1 - \mathrm{P}\big(T_1 \le k, T_2 \le k, \ldots, T_{10} \le k\big) \\ =& 1 - \big\{\mathrm{P}(T \le k)\big\}^{10} && \text{(Independency)} \\ =& 1 - \big\{1 - (1-p)^k\big\}^{10}.\end{aligned}$$

Question 5

Suppose $X \sim \text{Poisson}(\mu)$. According to Problem 3 from Section Sheet 13, we have

$$\mathrm{E}[X] = \mathrm{Var}(X) = \mu.$$

Question 5 - (i)

$$\mathrm{E}[X(X-1)] = \mathrm{E}[X^2] - \mathrm{E}[X] = \big\{\,\mathrm{Var}(X) + \mathrm{E}[X]^2\big\} - \mathrm{E}[X] = \mu + \mu^2 - \mu = \mu^2.$$

Question 5 - (ii)

$$\mathrm{E}[X^2] = \mathrm{Var}(X) + \mathrm{E}[X]^2 = \mu + \mu^2.$$

Question 5 - (iii)

$$\mathrm{E}[X(X+1)] = \mathrm{E}[X^2] + \mathrm{E}[X] = \big\{\,\mathrm{Var}(X) + \mathrm{E}[X]^2\big\} + \mathrm{E}[X] = \mu + \mu^2 + \mu = \mu^2 + 2\mu.$$

Question 6

This situation can be modeled by a negative hypergeometric distribution since we are sampling without replacement and we're interested in the number of cards drawn until a certain number of successes (3 Aces) are obtained.

For this negative binomial distribution, the probability mass function (PMF) of drawing exactly x cards to get 3 Aces is:

$$\mathrm{P}(X = x) = \binom{x-1}{3-1}\Big(\frac{4}{52}\Big)^3\Big(1 - \frac{4}{52}\Big)^{x-3}$$

where

- $\frac{4}{52}$ is the probability of drawing an Ace (success),
- $1 - \frac{4}{52}$ is the probability of not drawing an Ace (failure).

Question 6 - (i)

$$\begin{aligned}\mathrm{P}(X > x) =& 1 - \mathrm{P}(X \leq x) \\ =& 1 - \sum_{k=3}^{x} \binom{k-1}{2} \Big(\frac{4}{52}\Big)^3 \Big(1 - \frac{4}{52}\Big)^{k-3}\end{aligned}$$

Question 6 - (ii)

$$\mathrm{P}(X = x) = \binom{x-1}{2} \Big(\frac{4}{52}\Big)^3 \Big(1 - \frac{4}{52}\Big)^{x-3}$$

Question 6 - (iii)

For a negative binomial distribution, the expected value of X (the number of trials until 3 successes) is given by $E[X] = \frac{r}{p}$, where $r = 3$ (the number of Aces we want) and $p = \frac{4}{52}$ (the probability of drawing an Ace on any given card). Therefore,

$$\mathrm{E}[X] = \frac{3}{\frac{4}{52}} = 3 \times \frac{52}{4} = 39.$$